

## 1.1

$(i, j) \in \Omega$ , where  $i, j \in \{1, 2, 3, 4, 5, 6\}$ .

Let  $A = \{\text{Second Roll is Higher than the first}\}$

| First Roll | P(A) |
|------------|------|
| 1          | 5/6  |
| 2          | 4/6  |
| 3          | 3/6  |
| 4          | 2/6  |
| 5          | 1/6  |
| 6          | 0/6  |

Thus  $|A| = 5 + 4 + 3 + 2 + 1 = 15$

$|\Omega| = 6 * 6 = 36$

$P(A) = \frac{|A|}{|\Omega|} = \frac{15}{36} = \frac{5}{12}$

## 1.5

**a**

$\Omega = \{1, 2, 3, 4, \dots, 40\}^5$  using the cartesian power

$\forall \omega \in \Omega P(\{\omega\}) = 1/40^5$

**b**

Let  $3E = \{3 \text{ of } 5 \text{ numbers are even } (1-40)\}$

Let  $E = 2, 4, 6, \dots, 40$

Let  $O = 1, 3, 5, \dots, 39$

Since  $|E| = |O|$  and  $E^c = O$  then  $P(E) = P(O) = 1/2$

This means we can then translate a set of 5 numbers to  $E, O^5$  resulting in  $2^5 = 32$  different  $E, O$  sets.

Therefore there are  $\binom{5}{3}$  combinations of 3 even numbers.  $\binom{5}{3} = \frac{5!}{3!(5-3)!} = \frac{5*4}{2} = 10$

Thus  $P(3E) = 10/32 = 5/16$

## 1.7

**a**

$$P(GYG) = \frac{3}{7} \frac{4}{6} \frac{2}{5} = \frac{4}{35}$$

**b**

$3P(GYG) = 12/35$  this is because the probability of any order of 2 green and one yellow is the same.

## 1.10

**a**

$$\Omega = \{\infty, 1, 2, 3, \dots\}$$

$$\forall \omega \in \Omega \ P(\omega) = \left(\frac{5}{6}\right)^{\omega-1} * \frac{1}{6}$$

**b**

$$P(\omega = \infty) = \lim_{\omega \rightarrow \infty} \left(\frac{5}{6}\right)^{\omega-1} * \frac{1}{6} = 0$$

$$P(\{\infty\}) = 0$$

## 1.12

**a**

$$P(\omega \leq 3) = P(1) + P(2) + P(3) = \frac{1}{6} + \frac{5}{36} + \frac{25}{216} = \frac{91}{216}$$

It is also important to note that  $P(1)$ ,  $P(2)$ ,  $P(3)$  are disjoint

**b**

$$P(E\omega) = 5/36 + 125/(216 * 6) + \dots$$

$$\text{Using geometric series sum: } P = \frac{5/36}{1-(5/6)^2} = \frac{5}{11}$$

## 1.20

**a**

$$\Omega = \{1, 2, 3, 4, 5, 6\}^4$$

$$\forall \omega \in \Omega P(\{\omega\}) = \frac{1}{|\Omega|} = \frac{1}{6^4}$$

**b**

$$|\Omega| = 6^4$$

$$B = \{\text{No 5's}\} \cup \{\text{Only 1 5}\}$$

$$|\{\text{No 5's}\}| = 5^4$$

$$|\{\text{Only 1 5}\}| = 4 * 5^3 \text{ (because the 5 can be in 4 different positions and other numbers are 5 options \{non 5\})}$$

$$|B| = 5^4 + 4 * 5^3$$

$$P(B) = \frac{5^4 + 4 * 5^3}{6^4} = \frac{1125}{6^4}$$

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$$A = \{\text{Only 5's}\} \cup \{\text{One non 5}\} \cup \{\text{2 5's}\}$$

$$|\{\text{Only 5's}\}| = 1$$

$$|\{\text{Only one non 5}\}| = 4 * 5 = 20 \text{ (non 5 can be in 4 different places with 5 different values)}$$

For the set of 2 5's, there are 6 different orientations of the numbers where  $x, y \in \{1, 2, 3, 4, 6\}$

$$xy55 \ x5y5 \ x55y \ 5xy5 \ 5x5y \ 55xy$$

$$\text{Thus } |\{2 \text{ 5's}\}| = 6 * 5^2 = 150$$

$$\text{Therefore } P(A) = \frac{150+20+1}{6^4} = \frac{171}{6^4}$$

**c**

$$A \cup B = \Omega$$

$$P(A) + P(B) = 1$$

$$\rightarrow \frac{171}{6^4} + \frac{1125}{6^4} = 1$$

$$\rightarrow \frac{1296}{6^4} = 1$$

$$\rightarrow \frac{1296}{1296} = 1$$

So the answer to part b holds true

## 1.22

**a**

$$\Omega = (i, j) | i \in \{\text{hearts, clubs, spades, diamonds}\}, j \in \{2, 3, 4, 5, 6, 7, 8, 9, 10, J, Q, K, A\}$$

$$\forall \omega \in \Omega \ P(\omega) = \frac{1}{|\Omega|} = \frac{1}{4 \cdot 13} = \frac{1}{52}$$

**b**

What is the probability of picking an Ace from the deck that is not the Ace of Spades?

**c**

Suppose \$ A \$ s.t.  $P(A) = \frac{1}{5}$

By fact 1.8, then  $P(A) = \frac{|A|}{|\Omega|} = \frac{1}{5}$

Plugging in our known values:  $\frac{|A|}{52} = \frac{1}{5}$

When solving for  $|A|$ :  $|A| = \frac{52}{5} = 10.4$

This is not possible because the size of a set cannot be a non whole number. Therefore  $\nexists A$  s.t.  $P(A) = \frac{1}{5}$

### 1.34

$$\Omega = (i, j) \forall i, j (0 \leq i \leq 1, 0 \leq j \leq 1)$$

Circle lies within the square iff  $(\frac{1}{3} < i < \frac{2}{3}) \wedge (\frac{1}{3} < j < \frac{2}{3})$

The range  $(\frac{1}{3}, \frac{2}{3})$  is one third the size of the possible range. Thus the probability the  $i$  is within the range is  $\frac{1}{3}$  and same for  $j$ .

Since we apply this range to both  $i$  and  $j$  the resulting probability is  $\frac{1}{3} \times \frac{1}{3} = \frac{1}{9}$